

# Lecture 1: Regression, identification, and the OLS estimator

Economics 527: Econometric Methods

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## Motivation and data

### A motivating example: Return to education

- Suppose wages are generated according to:

$$\ln \text{Wage} = \beta_1 + \beta_2 \text{Education} + \beta_3 \text{Gender} + \beta_4 \text{Race} \\ + \beta_5 \text{Urban} + \beta_6 \text{Experience} + U.$$

- We have a sample of workers and, for each worker, we observe Wage, Education, etc.
- $U$  collects everything besides Education, Gender, etc. that affects wages. Ability is first on that list, and it is unobserved (latent).
- The unknown coefficient  $\beta_2$  measures the return to education, holding everything else fixed.
- Questions/Issues:
  - Why in this equation  $\beta_2$  measures the return to education;
  - How to “learn”  $\beta_2$  from random data;
  - What data randomness means;
  - What has to be assumed about the unobserved  $U$  to be able to learn  $\beta_2$ .

## Levels and logs

- **Level-level**,  $y = \beta x$ :

$$\beta = \frac{dy}{dx},$$

and the proportional change in  $y$  per unit of  $x$  is  $\beta/y$ , which depends on where  $y$  is.

- **Log-level**,  $\ln y = \beta x$ :

$$\beta = \frac{d \ln y}{dx} = \frac{dy/dx}{y},$$

so  $\beta$  is the proportional rate of change of  $y$  with respect to  $x$ . Multiply by 100 for a percentage.

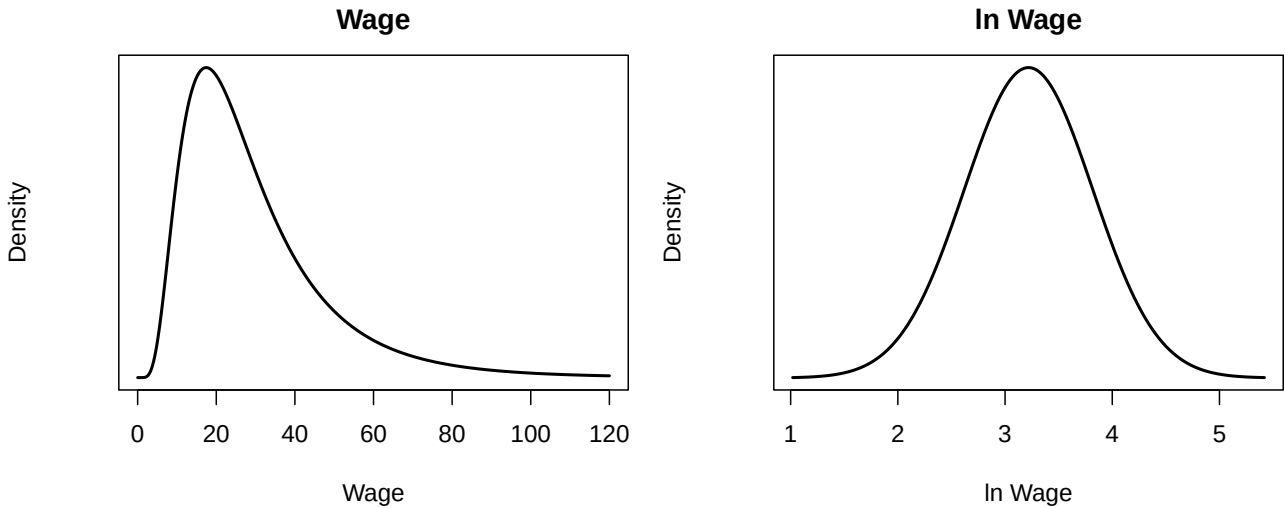
- **Log-log**,  $\ln y = \beta \ln x$ :

$$\beta = \frac{d \ln y}{d \ln x} = \frac{dy/y}{dx/x},$$

the elasticity of  $y$  with respect to  $x$ .

## Logs often “work better”

- Wages are bounded below by zero and have a long right tail. Their log is close to symmetric.



- A symmetric dependent variable is not required by anything that follows, but specifications with logs often “work better” in practice.

## The data

- We observe a **sample** of  $n$  workers:

$$\{(Y_i, X_i) : i = 1, \dots, n\}.$$

- For worker  $i$ ,  $Y_i = \ln \text{Wage}_i$  is a scalar ( $1 \times 1$ ) and  $X_i$  stacks the  $k$  regressors, with a 1 in the first position for the intercept:

$$X_i = \begin{pmatrix} 1 \\ \text{Educ}_i \\ \text{Gender}_i \\ \vdots \end{pmatrix} \quad (k \times 1), \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix} \quad (k \times 1).$$

- The wage equation for worker  $i$  is

$$Y_i = X_i^\top \beta + U_i = \sum_{j=1}^k X_{ij} \beta_j + U_i,$$

where  $X_i^\top \beta = \beta^\top X_i$  is a scalar.

## What is random and what is fixed

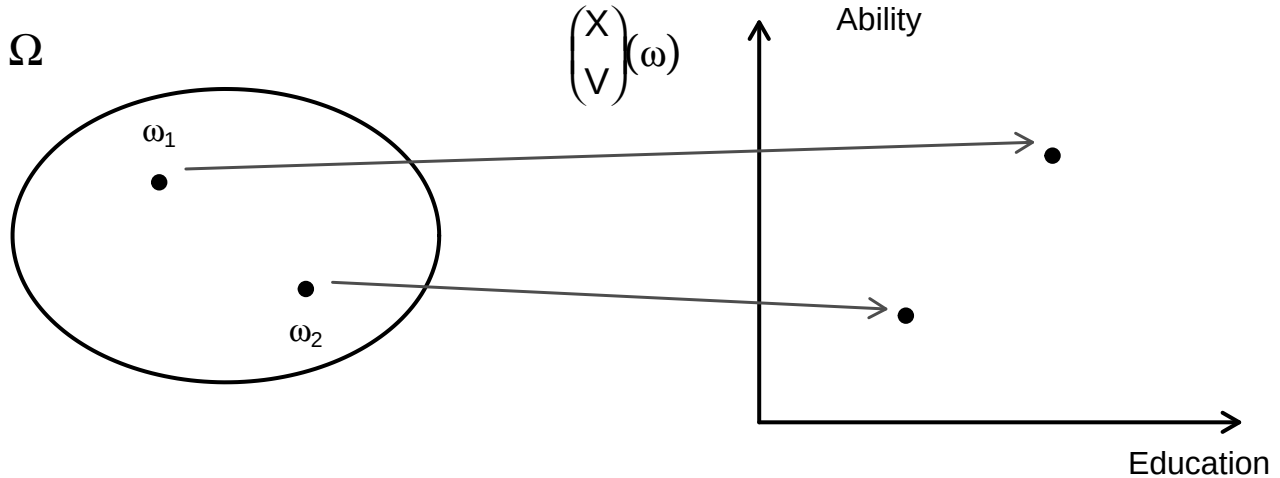
- $\beta$  is **fixed** and unknown.
- $X_i$ ,  $U_i$ , and therefore  $Y_i$  are **random**.
- The **population** is the distribution that generates  $\{(Y_i, X_i) : i = 1, \dots, n\}$ , a probability distribution rather than a group of people.
- In a cross-section, each observation is a different worker, so we take the observations to be drawn **independently and identically distributed**:

$$(Y_i, X_i) \text{ are iid, with joint distribution } F_{Y, X_1, \dots, X_k}.$$

- Note that independence is between  $(Y_i, X_i)$  and  $(Y_j, X_j)$  for  $i \neq j$ , not between  $Y_i$  and  $X_i$ .
- Under the iid assumption, we can take the population to be the joint distribution of one observation  $(Y_i, X_i)$ .

## Modelling the randomness

- A random experiment has a sample space  $\Omega$  of outcomes  $\omega$ . These are possible states of the world.
- Let  $X$  denote education and  $V$  ability.
- The pair  $(X, V)$  is a function from  $\Omega$  to  $\mathbb{R}^2$ : for the state of the world  $\omega$ ,  $X(\omega)$  is the education and  $V(\omega)$  is the ability.



- A probability  $P$  assigns to each event  $A$  a number  $P(A) \in [0, 1]$ . It induces a distribution for  $(X, V)$ .
- The joint CDF of  $(X, V)$  is a statement about sets of outcomes: for all  $x, v \in \mathbb{R}$ ,

$$\begin{aligned} F_{X,V}(x, v) &= P(X \leq x, V \leq v) \\ &= P(\{\omega \in \Omega : X(\omega) \leq x, V(\omega) \leq v\}). \end{aligned}$$

## Probability

### Random variables and the CDF

- A random variable is a map  $Y : \Omega \rightarrow \mathbb{R}$ . For a set  $A \subset \mathbb{R}$ ,

$$P(Y \in A) = P(\{\omega : Y(\omega) \in A\}).$$

- Taking  $A = (-\infty, y]$  gives the **cumulative distribution function**:

$$F_Y(y) = P(Y \leq y), \quad y \in \mathbb{R}.$$

- Properties:
  - $F_Y$  is nondecreasing and right-continuous,  $F_Y(y) \rightarrow 0$  as  $y \rightarrow -\infty$  and  $F_Y(y) \rightarrow 1$  as  $y \rightarrow \infty$ ;
  - $P(a < Y \leq b) = F_Y(b) - F_Y(a)$ ;
  - a jump of  $F_Y$  at  $y_1$  has size  $P(Y = y_1)$ ; where  $F_Y$  is continuous,  $P(Y = y_1) = 0$ .

### Continuous random variables and the PDF

- When  $F_Y$  is differentiable, the **probability density function** is

$$f_Y(y) = \frac{dF_Y(y)}{dy}, \quad \text{so} \quad F_Y(y) = \int_{-\infty}^y f_Y(v) dv.$$

- $f_Y(y) \geq 0$ ,  $\int_{-\infty}^{\infty} f_Y(y) dy = 1$ , and for  $A \subset \mathbb{R}$ ,

$$P(Y \in A) = \int_A f_Y(y) dy.$$

- A density is not a probability:  $P(Y = y) = 0$  for every  $y$ , and  $f_Y(y)$  can exceed one.
- Everything in this course is written for the continuous case. The discrete case replaces integrals with sums.

## Expectation

- For a function  $h$ ,

$$E[h(Y)] = \int_{-\infty}^{\infty} h(y) f_Y(y) dy.$$

- Throughout, every expectation we write down is assumed to exist and to be finite.
- $E[Y]$  is the first moment,  $E[Y^2]$  the second, and

$$\text{Var}(Y) = E[(Y - E[Y])^2] = E[Y^2] - (E[Y])^2.$$

- The mean is the best constant predictor of  $Y$  under squared loss:

$$E[(Y - c)^2] = E[(Y - E[Y])^2] + (E[Y] - c)^2 \geq \text{Var}(Y),$$

with equality only at  $c = E[Y]$ . (The cross term  $2(E[Y] - c)E[Y - E[Y]]$  is zero.)

- Under absolute loss the minimiser of  $E[|Y - c|]$  is the median instead. Which loss we use decides which feature of the distribution we end up estimating.

## Joint distributions

- For two random variables  $Y$  and  $X$  on the same  $\Omega$ :

$$F_{Y,X}(y, x) = P(Y \leq y, X \leq x),$$

$$f_{Y,X}(y, x) = \frac{\partial^2 F_{Y,X}(y, x)}{\partial y \partial x}.$$

- $f_{Y,X} \geq 0$ ,  $\iint f_{Y,X}(y, x) dy dx = 1$ , and for  $A \subset \mathbb{R}^2$ ,

$$P((Y, X) \in A) = \iint_A f_{Y,X}(y, x) dy dx.$$

- With  $k$  regressors the joint CDF of one observation is

$$F_{Y, X_1, \dots, X_k}(y, x_1, \dots, x_k) = P(Y \leq y, X_1 \leq x_1, \dots, X_k \leq x_k).$$

## From the joint density to a marginal

- **Claim.** The marginal density of  $X$  is obtained by integrating  $Y$  out:

$$f_X(x) = \int_{-\infty}^{\infty} f_{Y,X}(y, x) dy.$$

- **Proof.** Start from the marginal CDF and differentiate:

$$\begin{aligned} F_X(x) &= P(X \leq x) = P(X \leq x, Y < \infty) \\ &= \int_{-\infty}^x \int_{-\infty}^{\infty} f_{Y,X}(y, u) dy du, \\ f_X(x) &= \frac{dF_X(x)}{dx} = \frac{d}{dx} \left[ \int_{-\infty}^x \int_{-\infty}^{\infty} f_{Y,X}(y, u) dy du \right] \\ &= \int_{-\infty}^{\infty} f_{Y,X}(y, x) dy, \end{aligned}$$

using  $\frac{d}{dx} \int_{-\infty}^x h(u) du = h(x)$ .

- The converse fails: knowing  $f_X$  and  $f_Y$  does **not** determine  $f_{Y,X}$ . The marginals say nothing about how  $Y$  and  $X$  move together.

## Conditional density

- The density of  $Y$  **given**  $X = x$ , defined where  $f_X(x) > 0$ :

$$f_{Y|X}(y | x) = \frac{f_{Y,X}(y, x)}{f_X(x)}, \quad y \in \mathbb{R}.$$

- This has the same shape as conditional probability for events,  $P(A | B) = P(A \cap B)/P(B)$ : the joint, rescaled by the marginal density of what we condition on.
- It is a proper density in  $y$ :

$$\int_{-\infty}^{\infty} f_{Y|X}(y | x) dy = \frac{1}{f_X(x)} \underbrace{\int_{-\infty}^{\infty} f_{Y,X}(y, x) dy}_{= f_X(x)} = 1.$$

## Independence

- $Y$  and  $X$  are **independent** if conditioning changes nothing:

$$f_{Y|X}(y | x) = f_Y(y) \quad \text{for all } y \text{ and all } x \text{ with } f_X(x) > 0.$$

- Equivalently the joint factorises:

$$\begin{aligned} f_{Y,X}(y, x) &= f_Y(y) f_X(x) \quad \text{for all } x, y, \\ \text{or equivalently} \quad F_{Y,X}(y, x) &= F_Y(y) F_X(x) \quad \text{for all } x, y. \end{aligned}$$

- Independence is a strong statement: it involves the whole conditional distribution. Regression will need much less.

## Expectation of a function of two variables

- For a function  $h$  of both arguments,

$$E[h(Y, X)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(y, x) f_{Y,X}(y, x) dy dx.$$

- With  $h(y, x) = yx$  this is the joint moment  $E[YX]$ . Centring both arguments gives the **covariance**:

$$\begin{aligned} \text{Cov}(Y, X) &= E[(Y - E[Y])(X - E[X])] \\ &= E[YX] - E[Y] E[X]. \end{aligned}$$

- When  $\text{Var}(Y)$  and  $\text{Var}(X)$  are positive and finite, dividing by the two standard deviations removes the units and gives the correlation,

$$\text{Corr}(Y, X) = \frac{\text{Cov}(Y, X)}{\sqrt{\text{Var}(Y) \text{Var}(X)}} \in [-1, 1],$$

where the bound is the Cauchy-Schwarz inequality.

- $Y$  and  $X$  are **uncorrelated** if  $\text{Cov}(Y, X) = 0$ , equivalently  $E[YX] = E[Y] E[X]$ .
- Uncorrelatedness is a single restriction. It says the best straight-line predictor of  $Y$  from  $X$  under squared loss has zero slope, and it says nothing about any other way the two variables might be related.

## Some implications

- Under independence the expectation of a product of separate functions factorises:

$$\begin{aligned} \mathbb{E}[h_1(Y) h_2(X)] &= \iint h_1(y) h_2(x) \underbrace{f_{Y,X}(y, x)}_{= f_Y(y) f_X(x)} dy dx \\ &= \int h_1(y) f_Y(y) dy \cdot \int h_2(x) f_X(x) dx \\ &= \mathbb{E}[h_1(Y)] \mathbb{E}[h_2(X)]. \end{aligned}$$

- This holds for **every** pair of functions  $h_1$  and  $h_2$ , not just for the identity.
- Taking  $h_1(y) = y$  and  $h_2(x) = x$  gives  $\mathbb{E}[YX] = \mathbb{E}[Y]\mathbb{E}[X]$ , so  $\text{Cov}(Y, X) = 0$ : independence implies uncorrelatedness.
- The converse fails, and it fails for the reason above: uncorrelatedness is the factorisation for one pair of functions, whereas independence is the factorisation for all of them.

## Conditional expectation

### Conditional expectation: at a point

- Fix  $x$  with  $f_X(x) > 0$  and take the mean of the conditional density:

$$\mathbb{E}[Y \mid X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(y \mid x) dy.$$

This is a **number**.

- Repeat for every  $x$  with  $f_X(x) > 0$ . The result is a function of  $x$ ,

$$x \mapsto \mathbb{E}[Y \mid X = x] =: g(x),$$

the **conditional expectation function**.

### Conditional expectation: as a random variable

- Now plug the random variable  $X$  into  $g$ :

$$\mathbb{E}[Y \mid X] := g(X) = \int_{-\infty}^{\infty} y f_{Y|X}(y \mid X) dy.$$

This is a **random variable**: a different draw of  $X$  gives a different value.

- The distinction matters for every proof that follows.  $\mathbb{E}[Y \mid X = x]$  is a number indexed by  $x$ ;  $\mathbb{E}[Y \mid X]$  is the random variable obtained by evaluating that function at  $X$ .
- The same construction applied to the squared deviation gives the **conditional variance**,

$$\text{Var}(Y \mid X) = \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2 \mid X],$$

again a random variable, and a function of  $X$ .

- Since  $\mathbb{E}[Y \mid X]$  is random, it has a mean of its own. What is it?

## Law of iterated expectations

- Claim (LIE).**

$$\mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[Y].$$

- Averaging the conditional mean over the distribution of  $X$  gives back the unconditional mean.
- Read from right to left, it says any expectation can be computed in two stages: first conditional on  $X$ , then over  $X$ . Most proofs in this course use it that way.

## Law of iterated expectations: proof

- Write the outer expectation as an integral against  $f_X$ , then unpack  $g$ :

$$\begin{aligned} \mathbb{E}[\mathbb{E}[Y | X]] &= \int_{-\infty}^{\infty} \underbrace{\mathbb{E}[Y | X = x]}_{g(x)} f_X(x) dx \\ &= \int_{-\infty}^{\infty} \left[ \int_{-\infty}^{\infty} y f_{Y|X}(y | x) dy \right] f_X(x) dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \underbrace{f_{Y|X}(y | x) f_X(x)}_{= f_{Y,X}(y,x)} dy dx \\ &= \int_{-\infty}^{\infty} y \underbrace{\left[ \int_{-\infty}^{\infty} f_{Y,X}(y, x) dx \right]}_{= f_Y(y)} dy \\ &= \int_{-\infty}^{\infty} y f_Y(y) dy = \mathbb{E}[Y]. \end{aligned}$$

- The step from the third line to the fourth changes the order of integration.

## Taking out what is known

- Conditional on  $X$ , any function of  $X$  is a constant and comes out of the expectation:

$$\begin{aligned} \mathbb{E}[Y \cdot h(X) | X] &= \int_{-\infty}^{\infty} y h(X) f_{Y|X}(y | X) dy \\ &= h(X) \int_{-\infty}^{\infty} y f_{Y|X}(y | X) dy = h(X) \mathbb{E}[Y | X]. \end{aligned}$$

- Two special cases used constantly:

$$\mathbb{E}[h(X) | X] = h(X), \quad \mathbb{E}[c | X] = c.$$

## Question

- Let  $Y, X, Z$  be random variables. Which of the following equals  $\mathbb{E}[\mathbb{E}[Y | X, Z] | Z]$ ?
  - $\mathbb{E}[Y]$
  - $\mathbb{E}[Y | X]$
  - $\mathbb{E}[Y | Z]$
  - 0

## Mean independence

- $Y$  is **mean independent** of  $X$  if the conditional mean does not move with the conditioning value:

$$\mathbb{E}[Y | X = x] = c \quad \text{for all } x \text{ with } f_X(x) > 0,$$

where the constant  $c$  does not depend on  $x$ .

- Question.** The constant  $c$  does not depend on  $x$ . What must it be?

- 0
- $\mathbb{E}[X]$
- $\mathbb{E}[Y]$
- 42

- Mean independence is weaker than independence:

- Under mean independence,  $\mathbb{E}[Y | X]$  does not depend on  $X$ , but  $\mathbb{E}[Y^2 | X]$ ,  $\text{Var}(Y | X)$ , etc., can change with  $X$ .
- Under independence, the whole conditional distribution is constant in  $X$ , so  $\text{Var}(Y | X)$ , etc., are constant too.

## A zero conditional mean

- Suppose

$$E[U | X] = 0.$$

By the previous slide this is mean independence of  $U$  from  $X$ , in the case where the constant is zero.

- Two consequences, by the LIE and by taking out what is known:

$$E[U] = E \left[ \underbrace{E[U | X]}_{=0} \right] = 0,$$

$$E[UX] = E[E[UX | X]] = E \left[ X \underbrace{E[U | X]}_{=0} \right] = 0.$$

The second line takes  $X$  out of the inner expectation, which is allowed because  $X$  is known once we condition on it.

- Putting the two together,

$$\text{Cov}(U, X) = E[UX] - E[U] E[X] = 0 - 0 = 0,$$

so  $U$  and  $X$  are uncorrelated.

## Three concepts of unrelatedness

- Ordered from strongest to weakest:

$$\text{independence} \implies E[U | X] = E[U] \implies \text{Cov}(U, X) = 0,$$

and neither arrow reverses.

- Mean independence allows  $\text{Var}(U | X)$  to depend on  $X$ ; independence does not. A variable can have a conditional mean that never moves with  $X$  while its conditional variance does.
- Uncorrelatedness is weaker still:  $\text{Cov}(U, X) = 0$  is a single restriction, and it is compatible with a conditional mean  $E[U | X]$  that varies with  $X$ , because  $\text{Cov}(U, X) = \text{Cov}(E[U | X], X)$ .

## The conditional mean minimises mean squared error

- Among all functions  $m$  of  $X$ , the conditional mean is the best predictor of  $Y$  under squared loss. Condition on  $X$  first:

$$\begin{aligned} E[(Y - m(X))^2 | X] &= E \left[ (Y - E[Y | X])^2 | X \right] \\ &\quad + (E[Y | X] - m(X))^2, \end{aligned}$$

because the cross term is  $2(E[Y | X] - m(X)) E[Y - E[Y | X] | X] = 0$ .

- Now average over  $X$  with the LIE:

$$E[(Y - m(X))^2] \geq E \left[ (Y - E[Y | X])^2 \right],$$

with equality only when  $m(X) = E[Y | X]$  with probability one.

- This is the reason regression is about  $E[Y | X]$  and not some other feature of the conditional distribution. Change the loss and a different feature wins.

# The linear regression model

## The object of interest

- The relationship between the regressors and the dependent variable is summarised by the conditional expectation function:

$$E[Y_i | X_i] = E[Y_i | X_{i1}, \dots, X_{ik}],$$

for instance

$$E[\ln \text{Wage}_i | \text{Education}_i, \text{Experience}_i, \text{Race}_i, \text{Gender}_i, \dots].$$

- The practical problem: with data  $\{(Y_i, X_i) : i = 1, \dots, n\}$ , estimate  $E[Y_i | X_i]$ .
- Without restrictions this is a function of  $k$  arguments to be learned from  $n$  points. The **parametric** approach assumes the functional form and leaves a finite number of unknown constants.

## The linear regression model

- Assumption.** The conditional expectation is linear in the parameters:

$$E[Y_i | X_i] = \beta_1 X_{i1} + \dots + \beta_k X_{ik} = X_i^\top \beta,$$

where  $X_i = (X_{i1}, \dots, X_{ik})^\top$  and  $\beta = (\beta_1, \dots, \beta_k)^\top \in \mathbb{R}^k$ .

- Write  $E[Y_i | X_i = x] = x^\top \beta = \sum_{l=1}^k \beta_l x_l$ . Differentiating in  $x_j$ ,

$$\frac{\partial E[Y_i | X_i = x]}{\partial x_j} = \frac{\partial(\beta_1 x_1 + \dots + \beta_k x_k)}{\partial x_j} = \beta_j.$$

- So  $\beta_j$  is the change in the conditional mean of  $Y_i$  per unit change in  $X_{ij}$ , holding the other regressors fixed: a **marginal effect**.
- “Linear” refers to  $\beta$ . The regressors can be anything: squares, logs, products, indicator variables.

## The error term

- Define** the error as the deviation from the conditional mean:

$$U_i := Y_i - E[Y_i | X_i] = Y_i - X_i^\top \beta.$$

- Then, by construction,

$$\begin{aligned} Y_i &= X_i^\top \beta + U_i, \\ E[U_i | X_i] &= \underbrace{E[Y_i | X_i]}_{= X_i^\top \beta} - \underbrace{E[X_i^\top \beta | X_i]}_{= X_i^\top \beta} = 0. \end{aligned}$$

- $U_i$  is not observable: the conditional expectation function is unknown, so the deviation from it is unknown.
- If the pairs  $(Y_i, X_i)$  are iid, then so are the  $U_i$ , since each is the same function of its own pair.

## Two ways to write the same model

- Either start from the conditional mean and define the error,

$$\left. \begin{aligned} E[Y_i | X_i] &= X_i^\top \beta \\ U_i &:= Y_i - X_i^\top \beta \end{aligned} \right\} \implies E[U_i | X_i] = 0,$$

- or start from the equation and the restriction on the error,

$$\left. \begin{aligned} Y_i &= X_i^\top \beta + U_i \\ E[U_i | X_i] &= 0 \end{aligned} \right\} \implies E[Y_i | X_i] = X_i^\top \beta.$$

## The intercept

- Consider an example with a single regressor  $X_i^* \in \mathbb{R}$ .
- Suppose  $Y_i = \beta^* X_i^* + U_i^*$  with  $E[U_i^* | X_i^*] = \alpha \neq 0$ .
- We can add and subtract  $\alpha$  to get:

$$Y_i = \alpha + \beta^* X_i^* + \underbrace{(U_i^* - \alpha)}_{U_i}, \quad E[U_i | X_i^*] = 0.$$

- Define vectors  $X_i = (1, X_i^*)^\top$  and  $\beta = (\alpha, \beta^*)^\top$ . Then

$$Y_i = X_i^\top \beta + U_i, \quad E[U_i | X_i] = E[U_i | 1, X_i^*] = 0,$$

as conditioning on 1 in addition to  $X_i^*$  does not change the conditional expectation.

- Here  $\alpha$  is the intercept: a coefficient on a constant regressor (i.e., a regressor equal to 1 for all observations  $i$ ).
- We include the intercept to absorb constant but non-zero conditional means of the error. This does not change the slope coefficients.
- However, if the conditional mean of the error is not constant, i.e.,  $E[U_i^* | X_i^*] = \alpha + \gamma X_i^*$ , then

$$Y_i = \alpha + (\beta^* + \gamma) X_i^* + \underbrace{(U_i^* - \alpha - \gamma X_i^*)}_{U_i}, \quad E[U_i | X_i^*] = 0,$$

and the slope coefficient is affected.

## Identification

### Identification: the question

- Consider the model  $Y_i = X_i^\top \beta + U_i$  with  $E[U_i | X_i] = 0$ :
  - $Y_i, X_i$  are observed,
  - $U_i$  is unobserved,
  - $\beta \in \mathbb{R}^k$  is unknown.
- Suppose the joint distribution of the observables  $F_{Y, X_1, \dots, X_k}(y, x_1, \dots, x_k)$  is **known** (we are at the population level). You have no information about  $U_i$  beyond  $E[U_i | X_i] = 0$ .
- Can you recover  $\beta$ ?
- If the answer is no with the whole distribution in hand, no amount of data will help, as the joint distribution of the data does not pin down  $\beta$ .

### Identification: the definition

- **Definition.**  $\beta$  is **identified** if it is uniquely determined by the joint distribution of the observed variables  $(Y_i, X_{i1}, \dots, X_{ik})$ .
- Two things can go wrong:
  - the distribution of the observables is consistent with more than one value of  $\beta$ ;
  - it is consistent with none, because the model is wrong.
- We ask: what does  $E[U_i | X_i] = 0$  let us compute from the distribution of  $(Y_i, X_i)$ , and does it pin  $\beta$  down?

### What the model gives us

- By  $E[U_i | X_i] = 0$  and the LIE,  $E[X_i U_i] = 0$ .
- Rewrite  $U_i$  in terms of the observables and  $\beta$ :

$$U_i = Y_i - X_i^\top \beta$$

- Obtain the **population normal equations**:

$$0 = E[X_i U_i] = E[X_i(Y_i - X_i^\top \beta)].$$

- Multiply out, one step at a time, distribute the expectation:

$$\begin{aligned} 0 &= E[X_i U_i] = E[X_i(Y_i - X_i^\top \beta)] \\ &= E[X_i Y_i - X_i X_i^\top \beta] \\ &= E[X_i Y_i] - E[X_i X_i^\top \beta] \\ &= E[X_i Y_i] - E[X_i X_i^\top] \beta, \end{aligned}$$

the last step because  $\beta$  is a vector of constants and comes out of the expectation.

- Rearranging,

$$E[X_i X_i^\top] \beta = E[X_i Y_i].$$

- Knowns:  $E[X_i Y_i]$  ( $k \times 1$ ) and  $E[X_i X_i^\top]$  ( $k \times k$ ), both features of the distribution of the observables. Unknown:  $\beta$ .
- If  $E[X_i X_i^\top]$  is invertible, the system has exactly one solution. Whether it is invertible is the whole question.

## The second-moment matrix

- Write  $X_i$  as a column and  $X_i^\top$  as a row, and multiply:

$$\begin{aligned} X_i X_i^\top &= \begin{pmatrix} X_{i1} \\ X_{i2} \\ \vdots \\ X_{ik} \end{pmatrix} (X_{i1} \quad X_{i2} \quad \cdots \quad X_{ik}) \\ &= \begin{pmatrix} X_{i1}^2 & X_{i1}X_{i2} & \cdots & X_{i1}X_{ik} \\ X_{i2}X_{i1} & X_{i2}^2 & \cdots & X_{i2}X_{ik} \\ \vdots & \vdots & \ddots & \vdots \\ X_{ik}X_{i1} & X_{ik}X_{i2} & \cdots & X_{ik}^2 \end{pmatrix} \quad (k \times k). \end{aligned}$$

Entry  $(j, l)$  of the product is  $X_{ij}X_{il}$ .

- The expectation is taken entry by entry:

$$E[X_i X_i^\top] = E \begin{bmatrix} X_{i1}^2 & X_{i1}X_{i2} & \cdots & X_{i1}X_{ik} \\ X_{i2}X_{i1} & X_{i2}^2 & \cdots & X_{i2}X_{ik} \\ \vdots & \vdots & \ddots & \vdots \\ X_{ik}X_{i1} & X_{ik}X_{i2} & \cdots & X_{ik}^2 \end{bmatrix}.$$

- It is **symmetric**: using  $(AB)^\top = B^\top A^\top$ ,

$$(E[X_i X_i^\top])^\top = E[(X_i X_i^\top)^\top] = E[X_i X_i^\top].$$

## The second-moment matrix is positive semidefinite

- **Definitions.** A  $k \times k$  matrix  $A$  is **positive semidefinite** (p.s.d.) if  $c^\top A c \geq 0$  for all  $c \in \mathbb{R}^k$ , and **positive definite** (p.d.) if  $c^\top A c > 0$  for all  $c \neq 0$ .
- Take any  $c \in \mathbb{R}^k$  and define the scalar

$$W_i = X_i^\top c = c_1 X_{i1} + \cdots + c_k X_{ik}, \quad c^\top X_i = (X_i^\top c)^\top = W_i.$$

- Then the quadratic form is the expectation of a square:

$$c^\top \mathbb{E} [X_i X_i^\top] c = \mathbb{E} \left[ \underbrace{c^\top X_i}_{\tilde{W}_i} \underbrace{X_i^\top c}_{\tilde{W}_i} \right] = \mathbb{E} \left[ \underbrace{W_i^2}_{\geq 0} \right] \geq 0.$$

- So  $\mathbb{E} [X_i X_i^\top]$  is p.s.d. whatever the distribution of the regressors. Positive **definiteness** is the extra step, and it can fail.

### When positive definiteness fails

- Suppose some  $\tilde{c} \neq 0$  gives  $\tilde{c}^\top \mathbb{E} [X_i X_i^\top] \tilde{c} = 0$ . With  $\tilde{W}_i = X_i^\top \tilde{c}$ ,

$$\begin{aligned} 0 = \tilde{c}^\top \mathbb{E} [X_i X_i^\top] \tilde{c} &= \mathbb{E} [\tilde{W}_i^2] \iff \tilde{W}_i = 0 \text{ with probability } 1 \\ &\iff \boxed{\tilde{c}_1 X_{i1} + \dots + \tilde{c}_k X_{ik} = 0} \end{aligned}$$

- With  $\tilde{c} \neq 0$ , say  $\tilde{c}_1 \neq 0$ , one regressor equals, with probability one, an exact linear combination of the others:

$$X_{i1} = -\frac{\tilde{c}_2}{\tilde{c}_1} X_{i2} - \dots - \frac{\tilde{c}_k}{\tilde{c}_1} X_{ik}.$$

This is **perfect multicollinearity**.

### Positive definiteness and the rank condition

- Putting the two directions together,

$$\begin{aligned} \mathbb{E} [X_i X_i^\top] \text{ is p.d.} &\iff X_i^\top c = 0 \text{ only if } c = 0 \\ &\iff \mathbb{E} [X_i X_i^\top] c = 0 \text{ only if } c = 0 \\ &\iff \text{rank} (\mathbb{E} [X_i X_i^\top]) = k. \end{aligned}$$

- Ruled out: A relation  $c_1 X_{i1} + c_2 X_{i2} = 0$  for some  $(c_1, c_2) \neq (0, 0)$ .
- $X_{i2} = X_{i1}^2$  is not linear and is fine, unless  $X_{i1}$  is binary, when  $X_{i1}^2 = X_{i1}$ .

### Identification: the result

- **Assume**
  1.  $Y_i = X_i^\top \beta + U_i$  with  $\beta \in \mathbb{R}^k$ ;
  2.  $\mathbb{E} [U_i | X_i] = 0$ ;
  3. No perfect multicollinearity:  $\text{rank} (\mathbb{E} [X_i X_i^\top]) = k$ .
- Then  $\beta$  is identified:
  - Assumption 1 and 2 produce:  $\mathbb{E} [X_i (Y_i - X_i^\top \beta)] = 0$ .
  - This implies  $\mathbb{E} [X_i X_i^\top] \beta = \mathbb{E} [X_i Y_i]$ .
  - Assumption 3 makes  $\mathbb{E} [X_i X_i^\top]$  invertible, so

$$\boxed{\beta = (\mathbb{E} [X_i X_i^\top])^{-1} \mathbb{E} [X_i Y_i]}$$

## Estimation and least squares

### From identification to an estimator

- Identification wrote  $\beta$  as a function of population moments:

$$\beta = (\mathbb{E} [X_i X_i^\top])^{-1} \mathbb{E} [X_i Y_i].$$

- **Method of moments.** Replace each expectation  $E[\cdot]$  by the sample average  $\frac{1}{n} \sum_{i=1}^n (\cdot)$ :

$$\hat{\beta} = \left( \frac{1}{n} \sum_{i=1}^n X_i X_i^\top \right)^{-1} \frac{1}{n} \sum_{i=1}^n X_i Y_i = \left( \sum_{i=1}^n X_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Y_i.$$

The two factors of  $\frac{1}{n}$  cancel.

- An **estimator** is any function of the sample  $\{(Y_i, X_i)\}$ . It may not depend on  $U_i$  or on  $\beta$ , which are unobserved.

### The same estimator from the moment condition

- Alternatively, start from  $E[X_i(Y_i - X_i^\top \beta)] = 0$  and replace the expectation by an average, evaluated at the estimator:

$$\frac{1}{n} \sum_{i=1}^n X_i (\underbrace{Y_i - X_i^\top \hat{\beta}}_{=: \hat{U}_i}) = 0 \quad \Rightarrow \quad \sum_{i=1}^n X_i X_i^\top \hat{\beta} = \sum_{i=1}^n X_i Y_i,$$

which is the same  $\hat{\beta}$ .

- $\hat{U}_i = Y_i - X_i^\top \hat{\beta}$  is the **residual**: the sample counterpart of the error  $U_i = Y_i - X_i^\top \beta$ . Unlike  $U_i$ , it is computable.
- $\hat{\beta}$  is built to make the residuals satisfy

$$\frac{1}{n} \sum_{i=1}^n X_i \hat{U}_i = 0,$$

which is the sample version of  $E[X_i U_i] = 0$ .

### The estimator is not the parameter

- Sample averages are random; expectations are fixed numbers. In general

$$\frac{1}{n} \sum_{i=1}^n X_i X_i^\top \neq E[X_i X_i^\top], \quad \frac{1}{n} \sum_{i=1}^n X_i Y_i \neq E[X_i Y_i],$$

and so in general  $\hat{\beta} \neq \beta$ .

- Flip a fair coin ten times: the sample proportion of heads is rarely one half, although the population proportion is exactly one half.
- If the errors  $U_1, \dots, U_n$  are continuously distributed,  $P(\hat{\beta} = \beta) = 0$ .
- $\hat{\beta}$  is a random vector: a different sample gives a different value. It has a distribution, and the next lecture is about where that distribution sits relative to  $\beta$  and how spread out it is.

### Sample orthogonality

- Since  $\hat{\beta}$  solves  $\sum_i X_i(Y_i - X_i^\top \hat{\beta}) = 0$ ,

$$\sum_{i=1}^n X_i \hat{U}_i = 0 \quad \Leftrightarrow \quad \begin{pmatrix} \sum_{i=1}^n X_{i1} \hat{U}_i \\ \vdots \\ \sum_{i=1}^n X_{ik} \hat{U}_i \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

- If the first regressor is the constant,  $X_{i1} = 1$ , the first equation says  $\sum_{i=1}^n \hat{U}_i = 0$ : the residuals average to zero.

## Matrix notation

- Stack the  $n$  observations:

$$Y_{n \times 1} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad U_{n \times 1} = \begin{pmatrix} U_1 \\ \vdots \\ U_n \end{pmatrix}, \quad X_{n \times k} = \begin{pmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{pmatrix}.$$

- Row  $i$  of  $X$  is the transpose of the regressor vector  $X_i$ ; column  $j$  of  $X$  holds regressor  $j$  for all  $n$  observations:

$$X = \begin{pmatrix} X_{11} & X_{12} & \cdots & X_{1k} \\ \vdots & \vdots & & \vdots \\ X_{n1} & X_{n2} & \cdots & X_{nk} \end{pmatrix}.$$

- The  $n$  equations  $Y_i = X_i^\top \beta + U_i$  become one:

$$Y = \begin{pmatrix} X_1^\top \beta \\ \vdots \\ X_n^\top \beta \end{pmatrix} + \begin{pmatrix} U_1 \\ \vdots \\ U_n \end{pmatrix} \iff Y_{n \times 1} = X_{n \times k} \beta_{k \times 1} + U_{n \times 1}.$$

## Two identities and the matrix form of the estimator

- Claim.**

$$X^\top Y = \sum_{i=1}^n X_i Y_i \quad (k \times 1), \quad X^\top X = \sum_{i=1}^n X_i X_i^\top \quad (k \times k).$$

- Proof of the second.** Transposing the stack of rows gives a row of columns:

$$\begin{aligned} X^\top X &= \begin{pmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{pmatrix}^\top \begin{pmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{pmatrix} = (X_1 \quad \cdots \quad X_n) \begin{pmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{pmatrix} \\ &= \underbrace{X_1 X_1^\top}_{k \times k} + \cdots + X_n X_n^\top = \sum_{i=1}^n X_i X_i^\top. \end{aligned}$$

The first identity is the same computation with the second factor  $X$  replaced by  $Y$ , whose  $i$ th row is the scalar  $Y_i$ .

- Substitute the two identities into the method-of-moments formula:

$$\begin{aligned} \hat{\beta}_{k \times 1} &= \left( \sum_{i=1}^n X_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Y_i \\ &= \left( X^\top X \right)^{-1}_{k \times k} X^\top Y_{k \times 1}. \end{aligned}$$

## Residuals in matrix form

- Residual vector and its defining property:

$$\begin{aligned} \hat{U} &= Y - X \hat{\beta}, \\ X^\top \hat{U} &= X^\top (Y - X \hat{\beta}) = X^\top Y - X^\top X \underbrace{(X^\top X)^{-1} X^\top Y}_{\hat{\beta}} \\ &= X^\top Y - X^\top Y = 0. \end{aligned}$$

## The sample rank condition

- $\text{rank}(X^\top X) = \text{rank}(X)$ .
- Since  $X^\top X$  is  $k \times k$ , its inverse exists exactly when  $\text{rank}(X) = k$ , meaning the columns of  $X$  are linearly independent.
- Let  $c \in \mathbb{R}^k$  and consider the linear combination of the columns of  $X$ :

$$Xc = c_1 \begin{pmatrix} X_{11} \\ \vdots \\ X_{n1} \end{pmatrix} + \cdots + c_k \begin{pmatrix} X_{1k} \\ \vdots \\ X_{nk} \end{pmatrix} = 0 \quad \text{only if } c = 0,$$

In particular  $n \geq k$ : at least as many observations as coefficients.

- What it means observation by observation:  $c_1 X_{i1} + \cdots + c_k X_{ik} = 0$  for all  $i = 1, \dots, n$  only if  $c = 0$ .
- But this is equivalent to the population condition  $\text{rank}(\mathbb{E}[X_i X_i^\top]) = k$ , when  $n \geq k$ .

## Least squares

- Take a candidate value  $b \in \mathbb{R}^k$  and measure how badly the fitted values  $X_i^\top b$  fit the data by the **sum of squared residuals**:

$$\begin{aligned} S(b) &= \sum_{i=1}^n (Y_i - X_i^\top b)^2 = \begin{pmatrix} Y_1 - X_1^\top b \\ \vdots \\ Y_n - X_n^\top b \end{pmatrix}^\top \begin{pmatrix} Y_1 - X_1^\top b \\ \vdots \\ Y_n - X_n^\top b \end{pmatrix} \\ &= (Y - Xb)^\top (Y - Xb) = \|Y - Xb\|^2, \end{aligned}$$

where for  $a \in \mathbb{R}^n$ ,  $\|a\|^2 = a^\top a = \sum_{i=1}^n a_i^2$ .

- The **ordinary least squares** estimator picks the  $b$  that fits best:

$$\hat{\beta}^{\text{OLS}} = \arg \min_{b \in \mathbb{R}^k} \sum_{i=1}^n (Y_i - X_i^\top b)^2 = \arg \min_{b \in \mathbb{R}^k} \|Y - Xb\|^2.$$

- **Claim.**  $\hat{\beta}^{\text{OLS}} = \hat{\beta} = (X^\top X)^{-1} X^\top Y$ : least squares and the method of moments deliver the same estimator.

## Proof: completing the square

- We will use  $(u + v)^\top (u + v) = u^\top u + v^\top v + 2u^\top v$ .
- Add and subtract  $X\hat{\beta}$  inside the criterion:

$$\begin{aligned} (Y - Xb)^\top (Y - Xb) &= (\underbrace{Y - X\hat{\beta}}_{\hat{U}} + X(\hat{\beta} - b))^\top (\hat{U} + X(\hat{\beta} - b)) \\ &= \hat{U}^\top \hat{U} + (\hat{\beta} - b)^\top X^\top X(\hat{\beta} - b) \\ &\quad + 2 \underbrace{\hat{U}^\top X}_{=(X^\top \hat{U})^\top = 0} (\hat{\beta} - b) \\ &= \underbrace{\hat{U}^\top \hat{U}}_{\text{free of } b} + (\hat{\beta} - b)^\top X^\top X(\hat{\beta} - b). \end{aligned}$$

- The cross term vanishes by the normal equations  $X^\top \hat{U} = 0$ . That is the only place the definition of  $\hat{\beta}$  enters.
- So minimising  $S(b)$  over  $b$  is the same as minimising the quadratic form  $(\hat{\beta} - b)^\top X^\top X(\hat{\beta} - b)$ .

## Proof: the quadratic form

- For any  $a \in \mathbb{R}^k$ , with  $z = Xa \in \mathbb{R}^n$ ,

$$a^\top X^\top X a = (Xa)^\top (Xa) = z^\top z = \sum_{i=1}^n z_i^2 \geq 0,$$

so  $X^\top X$  is p.s.d. It is also symmetric, since  $(X^\top X)^\top = X^\top X$ .

- Under the rank condition,  $Xa = 0$  only if  $a = 0$ , so  $\sum_i z_i^2 = 0$  only if  $a = 0$ :  $X^\top X$  is **positive definite**.
- We have:

$$S(b) = S(\hat{\beta}) + \underbrace{(\hat{\beta} - b)^\top X^\top X (\hat{\beta} - b)}_{\geq 0, \text{ and } = 0 \text{ iff } b = \hat{\beta}} \geq S(\hat{\beta}).$$

- Hence  $\hat{\beta} = (X^\top X)^{-1} X^\top Y$  is the unique minimiser of the sum of squared residuals: it **is** the OLS estimator.

## The same result from the first-order condition

- Expand the criterion,  $S(b) = Y^\top Y - 2b^\top X^\top Y + b^\top X^\top X b$ , and use  $\partial(x^\top c)/\partial x = c$  together with  $\partial(x^\top A x)/\partial x = 2Ax$  for symmetric  $A$ :

$$\frac{\partial S(b)}{\partial b} = -2X^\top Y + 2X^\top X b = 0 \implies b = (X^\top X)^{-1} X^\top Y.$$

- The Hessian  $\partial^2 S / \partial b \partial b^\top = 2X^\top X$  is positive definite under the rank condition, so this is a minimum.
- The first-order condition, rewritten as  $X^\top (Y - Xb) = 0$ , **is** the normal equations. Least squares and the method of moments are two routes to one estimator.

## Population least squares

- The population has its own least-squares problem, and  $\beta$  solves it.
- Claim.** Suppose  $Y_i = X_i^\top \beta + U_i$  with  $E[U_i | X_i] = 0$ , and  $\text{rank}(E[X_i X_i^\top]) = k$ . Then

$$\beta = \arg \min_{b \in \mathbb{R}^k} E[(Y_i - X_i^\top b)^2],$$

and the minimiser is unique.

- So OLS is the sample analogue of  $\beta$  in two ways: replace population moments with sample moments, or replace expected squared error with average squared error.

## Population least squares: proof

- Condition on  $X_i$ , use that the conditional mean minimises conditional mean squared error, then apply the LIE:

$$\begin{aligned} E[(Y_i - X_i^\top b)^2] &= E[E[(Y_i - X_i^\top b)^2 | X_i]] \\ &\geq E\left[E\left[(Y_i - \underbrace{E[Y_i | X_i]}_{X_i^\top \beta})^2 | X_i\right]\right] \\ &= E[(Y_i - X_i^\top \beta)^2]. \end{aligned}$$

- Equality at some  $b^*$  requires  $X_i^\top b^* = E[Y_i | X_i] = X_i^\top \beta$  with probability one, so  $X_i^\top (b^* - \beta) = 0$  with probability one, and with no multicollinearity  $b^* = \beta$ .

## Other linear estimators, for contrast

- Nothing forces the weights in the normal equations to be equal across observations. For a known symmetric positive definite  $n \times n$  matrix  $W$ ,

$$\tilde{\beta} = (X^\top W X)^{-1} X^\top W Y$$

is also a function of the sample, and  $W = I_n$  gives back  $\hat{\beta}$ .

- In general  $\tilde{\beta}$  minimises  $(Y - Xb)^\top W(Y - Xb)$ . With  $W$  diagonal,  $W = \text{diag}(w_1, \dots, w_n)$  and  $w_i > 0$ , this is **weighted least squares**: it minimises  $\sum_i w_i (Y_i - X_i^\top b)^2$ . Taking  $W$  to be the inverse of the variance matrix of the errors gives **generalised least squares**.
- Which of these is a good estimator, and in what sense, is the subject of the next lecture: bias, variance, and the Gauss-Markov theorem.

## Summary: the model and identification

- The object of interest is the conditional expectation  $E[Y_i | X_i]$ ; the linear model assumes  $E[Y_i | X_i] = X_i^\top \beta$ , which is the same as  $Y_i = X_i^\top \beta + U_i$  with  $E[U_i | X_i] = 0$ .
- Probability supplied what was needed on the way: densities and conditional densities, the law of iterated expectations, taking out what is known, and mean independence.
- **Identification.**  $E[U_i | X_i] = 0$  gives  $E[X_i U_i] = 0$ , hence the population normal equations  $E[X_i(Y_i - X_i^\top \beta)] = 0$ , that is  $E[X_i X_i^\top] \beta = E[X_i Y_i]$ . If  $E[X_i X_i^\top]$  is positive definite, which means no exact linear relation among the regressors, then

$$\beta = (E[X_i X_i^\top])^{-1} E[X_i Y_i].$$

## Summary: the estimator

- **Estimation.** Under the sample rank condition  $\text{rank}(X) = k$ , replacing expectations by sample averages gives

$$\hat{\beta} = \left( \sum_{i=1}^n X_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Y_i = (X^\top X)^{-1} X^\top Y,$$

which is also the unique minimiser of  $\sum_i (Y_i - X_i^\top b)^2$ : the OLS estimator.

- $\hat{\beta}$  solves the sample normal equations  $X^\top (Y - X\hat{\beta}) = 0$ , that is  $X^\top \hat{U} = 0$ : the sample counterpart of the population normal equations  $E[X_i(Y_i - X_i^\top \beta)] = 0$ .